

Additional Experiments

Frontier Transfer. We explore whether configurations that perform well on one task remain effective on other tasks. For each source task, we construct its task-specific Pareto frontier using 200 policy rollouts from a diffusion policy trained exclusively on that task. We then apply the configurations on this frontier to each of the other three tasks and retain the Pareto-optimal points to form a transferred frontier.

To measure how closely each transferred frontier matches the target task’s native frontier, we compute the hypervolume ratio (HVR). Specifically, we divide the area under the transferred frontier by the area under the native frontier. The trigger-time range extends from zero to the latest non-dominated point reached by either frontier. An HVR near one indicates that the transferred configurations retain most of the accuracy–latency tradeoff obtained by task-specific tuning.

The results for each source-target task pair are shown in Figure 6. Across the 12 ordered source-target pairs, Reward-Dagger achieves a mean HVR of 0.92 and a median HVR of 0.93, indicating that *high-performing regions of the gate configuration space are largely shared across tasks*.

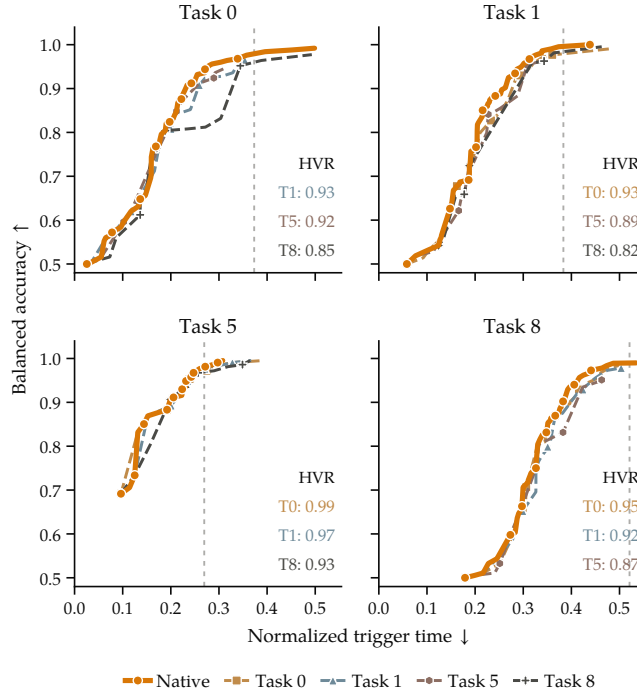


Fig. 6: **Frontier transfer of Reward-Dagger gate configurations.** Each panel corresponds to a target task. The solid orange curve denotes the native frontier obtained through task-specific tuning, and each dashed curve is obtained by applying configurations from another task’s frontier without retuning. Numbers indicate the hypervolume ratios (HVR) for each transferred curve. The dashed grey line indicates the mean normalized length of successful trajectories.

Different Reward Model Backbone in Real World Tasks. We further replace Robometer with RynnValue on the *Stack* and *Pick* policy-improvement experiments while directly reusing the Robometer-selected gating parameters, without additional tuning. As shown in Table V, RynnValue yields smaller gains in autonomous success but comparable ROHE overall, indicating that the proposed gating mechanism is not specific to a particular reward-model backbone.

TABLE V: **Different reward model backbone ablation on real-world tasks.**

Task	Method	Success rate (%)			ROHE		
		R0	R1	R2	R0	R1	R2
Stack	Robometer-4B	65	75	80	.667	.731	.643
	RynnValue-4B	65	65	70	.606	.645	.613
Pick	Robometer-4B	20	30	40	.541	.533	.588
	RynnValue-4B	20	25	25	.515	.556	.571

TABLE VI: Hyperparameter grids for detection frontier analysis

Method	Role	Parameter	Values	Grid size
	Frontier	FPR cap β	$\mathcal{B}$	22
Reward-DAGger	Inner search	Short-window length L_s	$\{20, 30, 50, 75, 100\}$	3,600
		Short-window threshold τ_s	$\{-0.9, -0.7, -0.5\}$	
		Minimum drop $\delta_{\min}$	$\{0, 0.05, 0.10, 0.20\}$	
		Long-window length L_l	$\{150, 175, \dots, 400, 450\}$	
		Long-window threshold τ_l	$\{-0.2, 0, 0.1, 0.3, 0.5\}$	
ThriftyDAGger	Frontier	Novelty/Q-risk quantile α	$\mathcal{A}$	43
Diff-DAGger	Frontier	Diffusion-loss quantile α	$\mathcal{A}$	43
UCF	Frontier	Uncertainty quantile α	$\mathcal{A}$	43
SAFE-LSTM	Frontier	Score quantile α	$\mathcal{A}$	43
logpZO	Frontier	Score quantile α	$\mathcal{A}$	43

$\mathcal{B} = \{0, 0.02\} \cup \{0.05, 0.10, \dots, 1.00\}$.
 $\mathcal{A} = \text{geospace}(10^{-4}, 0.5, 41) \cup \{10^{-3}, 10^{-2}\}$.

TABLE VII: Training and evaluation details

	Simulated tasks				Real-robot tasks			
	Task 0	Task 1	Task 4	Task 6	Stack	Pick	Put	Remove
Successful offline transitions	9,293	12,702	10,353	9,245	4,265	3,536	3,436	7,454
Observations	Agent-view RGB, wrist RGB, and proprioception				Exterior RGB, wrist RGB, joint positions (7), and gripper state (1)			
Image crop size	224 × 224 (resize with padding)							
Action chunk (predicted/executed)	20/10				15/8			
Control frequency	20 Hz				15 Hz			
Episode horizon	800 steps				900 steps			
Student policy	Diffusion policy trained on offline expert demonstrations				$\pi_{0.5}$ -DROID			
Data Aggregation	3 rounds, 3000 transitions per round				2 rounds, 1500 transitions per round			
Training per round	Full FT, 4k steps, batch size 256, constant lr 1e-4				LoRA, 3k steps, batch size 32, cosine lr 2.5×10^{-5}			
Expert	$\pi_{0.5}$ -LIBERO				Human teleoperator			
Evaluation rollouts per round	100				20			